

Pipeline Diagram

Cycle:	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21
Line 3 CMP:	IF	ID	EX	MEM	WB																
Line 4 BRGE:		IF	ID	**	EX	MEM	WB														
Line 5 LOAD:				IF	ID	EX	MEM	WB													
Line 6 SUB:					IF	ID	EX	MEM	WB												
Line 7 CMP:						IF	ID	EX	MEM	WB											
Line 8 BRGE:							IF	ID	**	EX	MEM	WB									
Line 9 LOADF:								IF	ID	EX	MEM	WB									
Line 10 LOADF:									IF	ID	EX	MEM	WB								
Line 11 CMP:										IF	ID	EX	MEM	WB							
Line 12 BRGE:											IF	ID	**	EX	MEM	WB					
Line 13 LOADF:												IF	ID	EX	MEM	WB					
Line 14 LOADF:													IF	ID	EX	MEM	WB				
Line 15 CMP:														IF	ID	EX	MEM	WB			
Line 16 JUMP:																IF	ID	EX	MEM	WB	

BRGE assumed not to be taken. If taken, Flush IF/ID of next instruction.

Hazard Table

Instruction pair	Hazard type	What causes it	Resolution
CMP → BRGE	Flags RAW	c14 writes flags, branch reads flags	Stall + forward
LOADF C → CMP	Load-use RAW	C written in WB, read in EX 2 cycles later	Forward MEM/WB → EX
BRGE taken	Control	Wrong instruction fetched after branch	Flush IF/ID
JUMP	Control	Wrong instruction fetched due to late knowing about JUMP	Flush IF/ID